PAPER_750: M51 Whirlpool and NGC 1316 Fornax A — MUGE with Simulation Scripts

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Framework: Universal Quantum Field Superconductive Framework (UQFF)
Session: 180 continuation | v5.38
Date: 2025
CP4 Class: #334 — M51NGC1316MUGESimulationCalculator

Abstract

Two Hubble-studied galaxies — M51 (the Whirlpool Galaxy, interacting with NGC 5195) and NGC 1316 (Fornax A, a post-merger elliptical) — represent opposite ends of the galaxy evolution spectrum. This paper derives tailored Master Universal Gravity Equations (MUGEs) for both systems using Hubble ACS datasets, incorporating interaction-specific environmental terms: F_tidal and _spiral for M51, and F_tidal + F_cluster + _dust for NGC 1316. Full Python simulation scripts (m51_simulation.py, ngc1316_simulation.py) are included for radial acceleration profile generation.

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1. Introduction

1.1 M51 (Whirlpool Galaxy)

M51 is a grand design spiral at 7.7 Mpc with an active tidal interaction companion NGC 5195. Hubble ACS observations (2005) provide: - M_visible = 1.2×10^{11} M - M_DM = 4×10^{1} M
- SFR 1 M /yr - M_BH 10 M - Distance = 7.7 Mpc, z 0.0015 - B 5×10 T

1.2 NGC 1316 (Fornax A)

NGC 1316 is a giant elliptical at 75 Mpc with a complex merger history: - M_total 5×10^{11} M (visible + DM) - M_DM = 1.5×10^{11} M - M_BH 10 M - Merger age: 1–3 Gyr ago (from ripples, dust lanes) - _dust 10 ^{21} kg/m^3

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2. M51 Whirlpool Galaxy MUGE

```
g_M51(r, t) = (G·M(t)) / (r(t)²) · (1+H(t,z)) · (1−B(t)/B_crit) · (1+F_env(t))
            + (U_g1 + U_g2 + U_g3' + U_g4) + U_i
            + (Λ·c²/3)
            + (ħ/√(Δx·Δp)) ·  ( _total·H· _total dV) · (2 /t_Hubble)
```

```
            +  _fluid·V·g
            + (M_vis + M_DM) · ( /  + (3·G·M)/r³)
```

M51 F_env Terms

```
F_env(t)_M51 = F_tidal +  _spiral_correction

F_tidal = G·M_NGC5195 / d_interaction²
  M_NGC5195   10¹  M  = 1.989×10   kg
  d_interaction   50 kpc = 1.54×10²¹ m
  F_tidal = 6.674×10 ¹¹ × 1.989×10   / (1.54×10²¹)²
  F_tidal   5.59×10 ¹³ m/s²  (normalized to F_env fraction)

 _spiral = A·e^(-r²/(2 ²))·e^(i(m· - ·t))
     1 kpc = 3.086×10¹  m  (spiral arm width)
  m = 2  (grand design two-arm spiral)
```

M51 H(t,z)

```
H(t,z) = H_0·√(0.3·(1+0.0015)³ + 0.7)   H_0·√0.7003   H_0
z = 0.0015 (negligible correction)
```

M51 Parameters Summary

Parameter	Value
M_visible	1.2×10^{11} M = 2.39×10^{1} kg
M_DM	4×10^{1} M = 7.96×10 kg
r_galaxy	30 kpc = 9.26×10^{2} m
B	5×10 T
SFR	1 M /yr = 6.3×10^{22} kg/yr
F_tidal	5.59×10^{13} m/s^2

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3. M51 Simulation Script (m51_simulation.py)

```python
import numpy as np
import matplotlib.pyplot as plt

# Constants
G = 6.6743e-11          # m^3 kg^-1 s^-2
H_0 = 70e3 / 3.086e22  # s^-1
Lambda = 1.1e-52         # m^-2
c = 3e8                  # m/s
hbar = 1.0546e-34        # J.s
```

```python
t_Hubble = 4.35e17        # s
B_crit = 1e11             # T
rho_vac_SCm = 7.09e-37    # J/m^3
rho_vac_UA = 7.09e-36     # J/m^3
mu_0 = 4 * np.pi * 1e-7   # H/m
M_sun = 1.989e30           # kg
kpc = 3.086e19             # m

# M51 Parameters
M_vis = 1.2e11 * M_sun
M_DM = 4.0e10 * M_sun
M_NGC5195 = 1e10 * M_sun
r_0 = 30e3 * kpc
d_inter = 50e3 * kpc
B = 5e-6                  # T
sigma_spiral = 1e3 * kpc
z = 0.0015
lambda_I = 1.0
F_RZ = 0.01
omega_i = 1.585e-8        # rad/s

def M_total():
    return M_vis + M_DM

def H_z(z):
    return H_0 * np.sqrt(0.3 * (1 + z)**3 + 0.7)

def F_tidal():
    return G * M_NGC5195 / d_inter**2

def U_i():
    return lambda_I * (rho_vac_SCm / rho_vac_UA) * omega_i * 1.0 * (1 + F_RZ)

def psi_spiral(r):
    A = 1e-10
    return A * np.exp(-r**2 / (2 * sigma_spiral**2))

def g_M51(r, t=0):
    M = M_total()
    H = H_z(z)
    F_env = F_tidal() / (G * M / r**2)  # normalized
    g_grav = (G * M / r**2) * (1 + H) * (1 - B / B_crit) * (1 + F_env)
    quantum = hbar / np.sqrt(1e-10 * 1e-20) * psi_spiral(r)**2 * (2 * np.pi / t_Hubble)
    dm_term = (M_vis + M_DM) * (1e-5 + 3 * G * M / r**3)
    cosmological = Lambda * c**2 / 3
    return g_grav + U_i() + cosmological + quantum + dm_term
```

```python
# Compute radial profile
r_vals = np.linspace(1e3 * kpc, 30e3 * kpc, 200)
g_vals = [g_M51(r) for r in r_vals]

plt.figure(figsize=(10, 6))
plt.plot(r_vals / kpc, g_vals, 'b-', linewidth=2, label='M51 MUGE Acceleration')
plt.xlabel('Radius (kpc)')
plt.ylabel('g_M51 (m/s²)')
plt.title('M51 Whirlpool Galaxy - UQFF MUGE Radial Profile')
plt.legend()
plt.grid(True, alpha=0.3)
plt.savefig('m51_gravity_profile.png', dpi=150)
plt.close()
print("M51 simulation complete. Profile saved to m51_gravity_profile.png")
```

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4. NGC 1316 (Fornax A) MUGE

```
g_NGC1316(r, t) = (G·M(t)) / (r(t)²) · (1+H(t,z)) · (1-B(t)/B_crit) · (1+F_env(t))
               + (U_g1 + U_g2 + U_g3' + U_g4) + U_i
               + (Λ·c²/3)
               + (ħ/√(Δx·Δp)) · ( _total·H· _total dV) · (2 /t_Hubble)
               +  _dust·V·g
               + (M_vis + M_DM) · ( /  + (3·G·M)/r³)
```

NGC 1316 F_env Terms

```
F_env(t)_NGC1316 = F_tidal + F_cluster

F_tidal: tidal from past spiral galaxy mergers
  F_tidal = G·M_spiral / d_spiral²
  M_spiral   10¹  M  (remnant progenitor)
  d_spiral   50 kpc
  F_tidal   5.59×10 ¹³ m/s²

F_cluster: star cluster tidal disruption
  F_cluster = k_cluster · M_cluster
  k_cluster   10 ¹² N/M
  M_cluster   10  M
  F_cluster   10  m/s²

 _dust: dust lane fluid term
   _dust = 10 ²¹ kg/m³  (ACS-derived dust column density)
```

NGC 1316 Mass Evolution

```
M(t) = M_vis + M_DM + M_merger(t)
M_merger(t) = M_spiral·e^(-t/ _merge)
 _merge   1 Gyr = 3.156×10¹  s
```

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5. NGC 1316 Simulation Script (ngc1316_simulation.py)

```python
import numpy as np
import matplotlib.pyplot as plt

# Constants (same as M51)
G = 6.6743e-11
H_0 = 70e3 / 3.086e22
Lambda = 1.1e-52
c = 3e8
hbar = 1.0546e-34
t_Hubble = 4.35e17
B_crit = 1e11
rho_vac_SCm = 7.09e-37
mu_0 = 4 * np.pi * 1e-7
M_sun = 1.989e30
kpc = 3.086e19

# NGC 1316 Parameters
M_vis = 3.5e11 * M_sun
M_DM = 1.5e11 * M_sun
M_spiral = 1e10 * M_sun
M_BH = 1e8 * M_sun
tau = 3.156e16          # 1 Gyr merger decay
d_spiral = 50e3 * kpc
rho_dust = 1e-21        # kg/m^3
B = 1e-4                # T
z = 0.005               # NGC 1316 redshift
k_cluster = 1e-12       # N/M_sun
M_cluster = 1e6 * M_sun
sigma_dust = 2e3 * kpc

def M_total(t):
    return M_vis + M_DM + M_spiral * np.exp(-t / tau)

def H_z(z):
    return H_0 * np.sqrt(0.3 * (1 + z)**3 + 0.7)

def F_env(t):
```

```python
    F_tidal = G * M_spiral / d_spiral**2
    F_cluster = k_cluster * M_cluster
    return F_tidal + F_cluster

def U_i():
    rho_vac_UA = 7.09e-36
    return 1.0 * (rho_vac_SCm / rho_vac_UA) * 1e-8 * 1.0 * 1.01

def g_NGC1316(r, t=3.156e15):  # t=100 Myr default
    M = M_total(t)
    H = H_z(z)
    F = F_env(t) / (G * M / r**2)
    g_grav = (G * M / r**2) * (1 + H) * (1 - B / B_crit) * (1 + F)
    g_dust = rho_dust * (4/3 * np.pi * r**3) * G * M / r**2
    dm_term = (M_vis + M_DM) * (1e-5 + 3 * G * M / r**3)
    cosmological = Lambda * c**2 / 3
    return g_grav + U_i() + cosmological + g_dust + dm_term

# Compute
r_vals = np.linspace(1e3 * kpc, 100e3 * kpc, 100)
g_vals = [g_NGC1316(r) for r in r_vals]

plt.figure(figsize=(10, 6))
plt.plot(r_vals / kpc, g_vals, 'r-', linewidth=2, label='NGC 1316 MUGE Acceleration')
plt.xlabel('Radius (kpc)')
plt.ylabel('g_NGC1316 (m/s²)')
plt.title('NGC 1316 Fornax A - UQFF MUGE Radial Profile')
plt.legend()
plt.grid(True, alpha=0.3)
plt.savefig('ngc1316_gravity_profile.png', dpi=150)
plt.close()
print("NGC 1316 simulation complete. Profile saved to ngc1316_gravity_profile.png")
```

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6. Comparative Analysis

Property	M51 (Whirlpool)	NGC 1316 (Fornax A)
Type	Sc spiral	E/S0 elliptical
Distance	7.7 Mpc	75 Mpc
Interaction	Active (NGC 5195)	Past mergers (~1-3 Gyr)
Key term	F_tidal (live)	F_cluster + _dust
Star formation	SFR 1 M /yr	~0 (quenched)
AGN activity	Moderate	Strong (radio lobes)
Key physics	Density waves (_spiral)	Dust lane drag (_dust)

7. Astrophysical Significance

M51: The F_tidal term drives the prominent spiral arms visible in Hubble observations. The _spiral density wave modulation affects star formation efficiency across the disk. NGC 5195's tidal force creates the bridge structure and enhanced SFR in the outer arms.

NGC 1316: The F_cluster term explains the deficit of low-mass globular clusters in the inner region (disrupted by tidal forces during multiple mergers). The _dust dust lane creates a complex hydrodynamic environment near the AGN.

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8. Conclusion

The M51 and NGC 1316 MUGEs demonstrate the UQFF framework's ability to model both active interaction (M51) and post-merger (NGC 1316) galaxy evolution with Hubble-calibrated parameters. The included Python simulation scripts provide quantitative radial acceleration profiles ready for comparison with observational rotation curves and X-ray surface brightness profiles. These two contrasting systems validate the modularity of F_env(t) across diverse galactic environments.
